

Vol 13 Chapter 3 — Prebiotic Forcing

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 3 — Prebiotic Forcing

Level 1 — one sentence

Prebiotic chemistry — the origin of life from non-living matter ~4 billion years ago — involved spontaneous formation of amino acids, nucleotides, lipids from simple precursors under early-Earth conditions, with substrate dynamics forcing chemistry toward configurations that support self-replication.

Level 2 — graduate-physicist precision

3.1 Miller-Urey experiment

Miller-Urey 1953: electric discharge in $\text{CH}_4 + \text{NH}_3 + \text{H}_2 + \text{H}_2\text{O} \rightarrow$ amino acids (glycine, alanine, aspartate, glutamate, etc.).

Demonstrates: building blocks of life form spontaneously under plausible early-Earth conditions.

Extended: nucleobases, sugars, lipids all synthesizable from prebiotic precursors (Sutherland 2009 nucleotide synthesis; Deamer membrane formation).

3.2 Early Earth conditions

~4.0-3.8 Gya: - Surface temperature: high (post-Late-Heavy-Bombardment cooling) - Atmosphere: reducing (CH_4 , NH_3 , H_2O , CO_2 ; little O_2) - Lightning, UV flux, hydrothermal vents - Liquid water ocean

First fossil evidence of life: ~3.5 Gya (stromatolites, Pilbara). Bracket: life arose 3.5-4.0 Gya, within ~500 Myr of Earth's surface stabilization.

3.3 RNA world hypothesis

Modern dogma (Watson-Crick): $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{protein}$. Origin problem: which came first?

RNA world (Gilbert 1986): RNA both encodes information AND catalyzes reactions (ribozymes). RNA is plausible first replicator.

Evidence: ribosome is a ribozyme (RNA catalyzes peptide bond formation); many ribozymes still functional in modern cells.

3.4 Substrate-forcing

BST view: substrate dynamics force chemistry toward configurations supporting Darwinian replicators.

Mechanism: substrate K-type configurations have natural attractors at: - Self-template structures (nucleotide pairing — Watson-Crick H-bonds) - Compartments (amphiphilic lipid bilayers) - Catalysts (folded macromolecules with active sites)

These substrate-natural attractors get visited by chemistry under sufficient time + energy throughput. Result: life is substrate-forced under early-Earth conditions, not coincidence.

3.5 Implications

If substrate-forced: life should arise wherever chemistry has enough time + energy (predicts: life common in the universe, possibly different substrate-K-type variants).

Falsifier: find planet with comparable conditions, age, chemistry, but no life. (Hard to falsify; Fermi paradox.)

3.6 K-audit anchors

- Vol 12 Ch 4: thermodynamics/kinetics
- Vol 6 (entropy)
- Casey's Principle (force from counting)

Level 3 — 5th-grader accessibility

Prebiotic chemistry = chemistry before life. Miller-Urey 1953: electric sparks in early-Earth-like gases → amino acids. RNA world: RNA can both store info and catalyze reactions; probably came before DNA + protein. Substrate-forcing: BST says the substrate's natural K-type attractors include self-replicating structures; given time + energy, life arises. Predicts: life common in universe. Falsifier: identical-conditions planet without life.

What comes next

Chapter 4 develops DNA and proton siblings.

Where to look this up

- Miller 1953
- Gilbert 1986 (RNA world)
- BST: Vol 12, Vol 6